

Exp 4: Data modelling and form design for a business application in Zoho Creator.

Register Number: 241401127

1. Aim

To design and build a functional "Bank Loan Application Form" application in Zoho Creator that demonstrates the correct usage of all thirteen Basic Fields available in the Form Builder, while solving a real banking requirement of collecting and pre-screening loan applications submitted by prospective borrowers.

2. Procedure

2.1 Problem Statement & Application Scope

A bank branch currently accepts loan applications through paper forms filled at the counter, leading to illegible handwriting, incomplete income details, and delays in routing applications to the correct loan officer based on loan type. There is no digital pre-screening step before the applicant is called in for document verification.

Design a low-code application using Zoho Creator that allows a prospective borrower to submit a loan application enquiry through a single online form. The form must capture personal identity, contact details, employment and income details, loan requirements, existing liabilities, and consent for a credit check — using every Basic Field type provided by Zoho Creator's Form Builder.

- Application Name: Loan Application & Pre-Screening Management
- Form Name: Loan_Application_Form
- Primary Users: Prospective borrowers / Bank loan officers
- Report View: Loan_Application_Form Report (list of all submitted loan applications)
- Objective of the Report: Enable loan officers to filter, pre-screen, and route applications for document verification and approval

2.2 Data Model & ER Diagram

A Data Model consists of both the visual Entity-Relationship (ER) diagram showing schema relationships and the structured Data Dictionary table defining field types and constraints. The structure below illustrates the internal schema generated by Zoho Creator for Loan_Application_Form, showing the primary form entity, composite data structures, system metadata fields, and choice lookup sets.

APPLICANT_INFO (Composite)

- First_Name : String
- Last_Name : String
- Email_Address : Email
- Contact_Number : Phone
- Residential_Address : Address

LOAN_REQUEST (Entity)

- [PK] PAN_Number : String
- Loan_Type : Dropdown
- Monthly_Income : Number
- Purpose_Of_Loan : Text

LOAN_APPLICATION_FORM (Form Entity)

- [PK] ID : BigInt (Auto-Generated)

- Applicant_Name : Name [Composite]
- Email_Address : Email (RFC 5322)
- Contact_Number : Phone (+91)
- Residential_Address : Address [Comp]
- PAN_Number : Text [Single]
- Date_Of_Birth : Date
- Employment_Type : Radio [Enum]
- Monthly_Income : Number
- Existing_Liabilities : List [Multi]
- Loan_Type : Picklist [Enum]
- Purpose_Of_Loan : TextArea
- Preferred_Callback_Time : Time
- Consent_For_Credit_Check : Checkbox [Bool]

EMPLOYMENT_TYPE_CHOICES : Salaried | Self-Employed

LOAN_TYPE_CHOICES : Home Loan | Personal Loan | Business Loan | Auto Loan | Education Loan

EXISTING_LIABILITIES_CHOICES : Car Loan | Personal Loan | Credit Card Dues | Education Loan | None of the above

Figure 1: Entity-Relationship (ER) Schema Structure for Loan Application Form in Zoho Creator

2.3 Basic Fields Mapping & Data Dictionary

Table 1 presents the mapping of all thirteen Basic Fields from the Zoho Creator Form Builder palette to their specific roles, purpose, and sample values in Loan_Application_Form.

Basic Field	Form Field Name	Field Link Name	Purpose & Validation Constraints	Sample Value
Name	Applicant Name	Applicant_Name	Captures full name of applicant using First/Last name sub-fields (Mandatory).	Karthick Raja
Email	Email Address	Email_Address	Captures a validated email address for sending application status updates.	karthik.raja@example.com
Address	Residential Address	Residential_Address	Captures the applicant's full residential address with city, state, and pincode.	7, Race Course Road, Coimbatore, TN, 641018
Phone	Contact Number	Contact_Number	Captures a validated mobile number for the bank to call and verify details.	+91 89440 33221
Single Line	PAN Number	PAN_Number	Captures a short free-text entry for the applicant's PAN.	EFGHI6789M
Multi Line	Purpose of Loan	Purpose_Of_Loan	Captures free-text description of the intended use of the loan amount.	Purchase of a new two-wheeler for daily commute.
Number	Monthly Income	Monthly_Income	Captures numeric value for gross monthly income. Constraint: must be > 0.	47000
Date	Date of Birth	Date_Of_Birth	Captures the applicant's date of birth using a date picker; used	27-Jun-1985

Basic Field	Form Field Name	Field Link Name	Purpose & Validation Constraints	Sample Value
			for age/eligibility checks (18+).	
Time	Preferred Callback Time	Preferred_Callback_Time	Captures the applicant's preferred time slot for a callback from the loan officer.	10:30 AM
Drop Down	Loan Type	Loan_Type	Presents a single-select list of loan products offered by the bank.	Auto Loan
Radio	Employment Type	Employment_Type	Captures a single mandatory selection among mutually exclusive employment categories.	Salaried
Multi Select	Existing Liabilities	Existing_Liabilities	Allows selection of one or more existing loans or financial commitments held by the applicant.	None of the above
Checkbox	Consent for Credit Check	Consent_For_Credit_Check	Captures the applicant's consent for the bank to perform a credit bureau check.	Checked (Agreed)

Table 1: Mapping and Specification of Thirteen Basic Fields in Loan_Application_Form

2.4 Form Layout & Configuration Steps

The Loan_Application_Form was organized into four logical, applicant-friendly sections:

- Section A — Personal Details: Applicant Name (Name), Date of Birth (Date), PAN Number (Single Line).
- Section B — Contact Details: Email Address (Email), Contact Number (Phone), Residential Address (Address).
- Section C — Income & Employment Details: Employment Type (Radio), Monthly Income (Number), Existing Liabilities (Multi Select).
- Section D — Loan Details & Consent: Loan Type (Drop Down), Purpose of Loan (Multi Line), Preferred Callback Time (Time), Consent for Credit Check (Checkbox).

The screenshot displays the Zoho Creator Form Builder interface. On the left, a 'Basic Fields' palette lists various field types like Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox. The main workspace shows a form layout with four sections: A - Personal Details, B - Contact Details, C - Income & Employment Details, and D - Loan Details & Consent. The 'Field Properties' inspector on the right is configured for the 'Applicant_Name' field, showing settings for field name, link name, validation (Mandatory), display fields (Last Name), data privacy (Contains personal data), data security (Data encryption is not supported), visibility (Show field to Everyone), and appearance (Field type: Name, Field size: Medium).

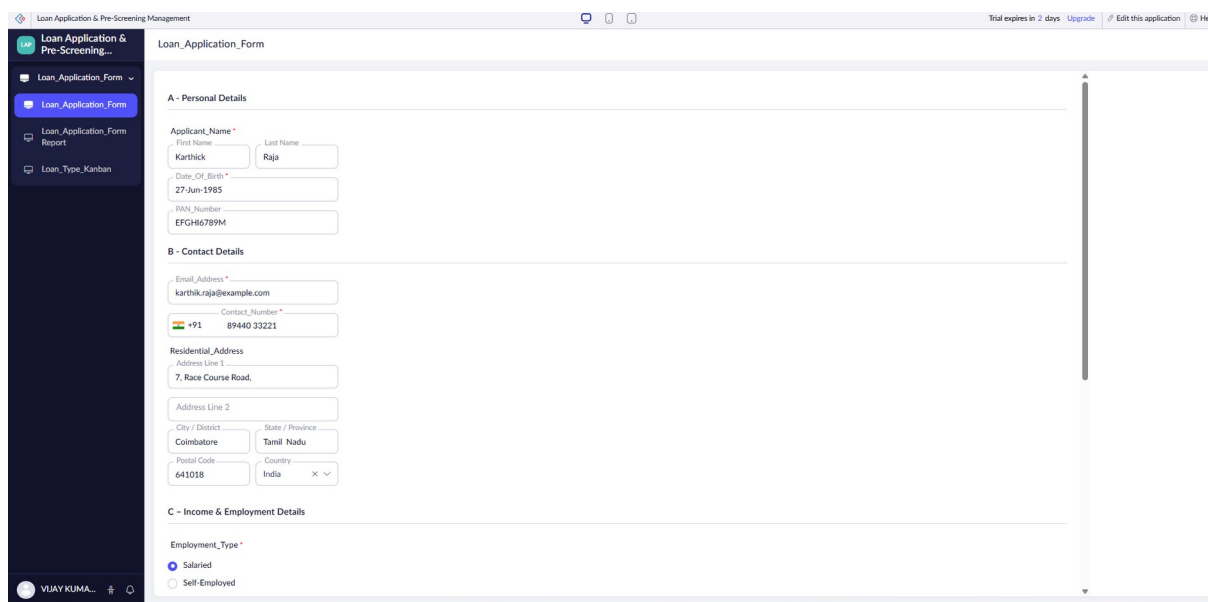
Figure 2: Form Builder Workspace & Field Properties Inspector Setup in Zoho Creator (all 13 Basic Fields)

2.5 Field-Level Validation Configuration

Field-level validations were configured directly on the form under Field Properties > Validation to enforce business logic checks: Applicant Name, Contact Number, Date of Birth, Employment Type, Loan Type, Monthly Income, and the Consent Checkbox were set as mandatory; the Email field's built-in RFC 5322 format validation was enabled; the Phone field's Indian country-code format was enabled; the Monthly Income field was restricted to values greater than 0; the Date of Birth field was restricted so the applicant must be at least 18 years old; and the form was configured so it cannot be submitted unless the credit-check consent checkbox is checked.

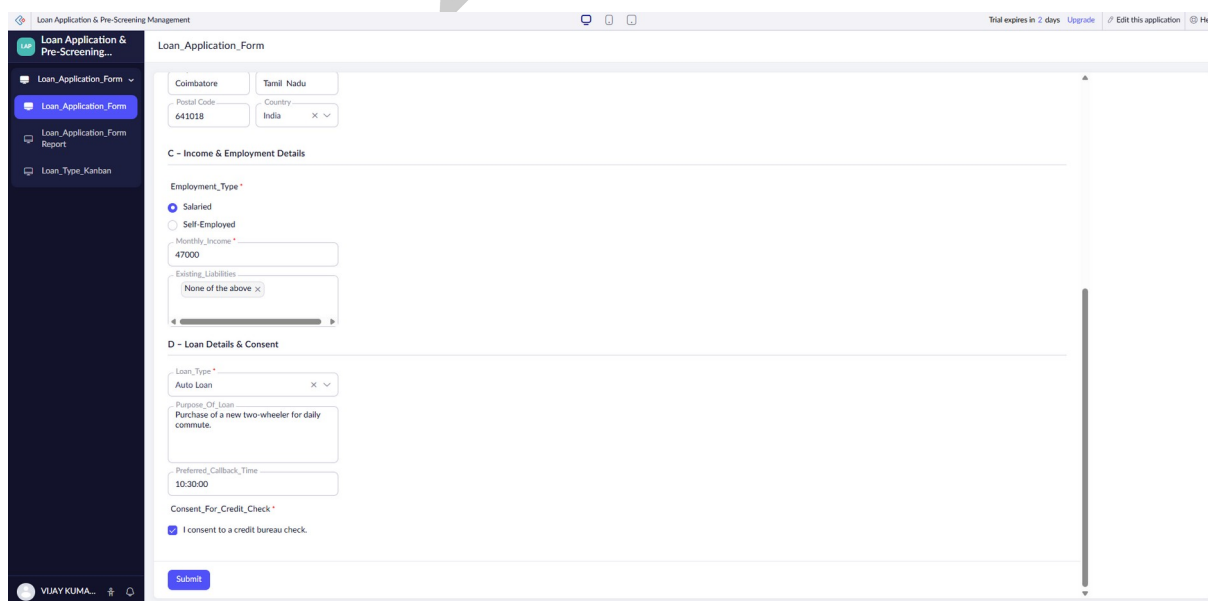
3. Output

The following figures show the live preview of the form (across its two scrollable sections) and the submitted applicant records in the report view:



The screenshot displays the 'Loan_Application_Form' in a web browser. The form is divided into three main sections: A - Personal Details, B - Contact Details, and C - Income & Employment Details. Section A includes fields for Applicant Name (First Name: Karthick, Last Name: Raja), Date of Birth (27-Jun-1985), and PAN Number (EFGH6789M). Section B includes Email Address (karthik.raja@example.com), Contact Number (+91 89440 33221), Residential Address (7, Race Course Road), City/District (Coimbatore), State/Province (Tamil Nadu), Postal Code (641018), and Country (India). Section C includes Employment Type (Salaried) and a radio button for Self-Employed. The form is part of a 'Loan Application & Pre-Screening' application, with a trial period of 2 days and options to upgrade, edit, or help.

Figure 3: Live Application Runtime Preview — Sections A & B (Personal and Contact Details)



The screenshot displays the 'Loan_Application_Form' in a web browser, showing sections C and D. Section C - Income & Employment Details includes Employment Type (Salaried), Monthly Income (47000), Existing Liabilities (None of the above), and a slider for Monthly Income. Section D - Loan Details & Consent includes Loan Type (Auto Loan), Purpose of Loan (Purchase of a new two-wheeler for daily commute), Preferred Callback Time (10:30:00), and Consent For Credit Check (I consent to a credit bureau check). The form is part of a 'Loan Application & Pre-Screening' application, with a trial period of 2 days and options to upgrade, edit, or help.

Figure 4: Live Application Runtime Preview — Sections C & D (Income, Loan Details & Consent)

Loan_Application_Form Report							
Applicant_Name	Date_Of_Birth	PAN_Number	Email_Address	Contact_Number	Residential_Address	Employment_Type	
Karthick Raja	27-Jun-1985	EFGH6789M	karthik.raja@example.com	+918944033221	7, Race Course Road, Coimbatore, Tamil Nadu, 641018, India	Salaried	
Meena Subramaniam	09-Mar-1992	DXNQR3345L	meena.subramaniam@example.com	+919655210987	22, Kamaraj Nagar, Madurai, Tamil Nadu, 625001, India	Self-Employed	
Arvind Kumar	03-Nov-1988	CMNOP9012Q	arvind.kumar@example.com	+919003188452	9, Gandhi Road, Coimbatore, Tamil Nadu, 641001, India	Salaried	
Priya Natarajan	14-Aug-1995	BPXTP5678K	priya.natarajan@example.com	+919876543210	45, Anna Nagar 2nd Street, Chennai, Tamil Nadu, 600040, India	Self-Employed	
Suresh Raghavan	22-Jan-1990	ABCDE1234F	suresh.raghavan@example.com	+919443022110	18, Lake View Road, Trichy, TN, 620001	Salaried	

Figure 5: Loan_Application_Form Report View Displaying Submitted Applicant Records

4. Result

The "Bank Loan Application Form" application was successfully built in Zoho Creator using all thirteen Basic Fields, with full field validations, section layout formatting, ER schema modeling, and report view generation meeting all assignment requirements.